

The constant-Gaussian fluctuation model: the finiteness threshold

Claude, with Daniel Murfet

2026-09-07

Abstract

Second half of the `rem:pop_vs_emp` dichotomy. For $X \sim N(0, v)$ the expected Gaussian moment $\mathbb{E}_+[J_p(X)] = \int_0^\infty s^{p-1} e^{-(\beta - \beta^2 v/2)s^2} ds$ (unit 161) is $J_p(0)(1 - \beta v/2)^{-p/2} < \infty$ when $\beta v < 2$ (`lintegral_gaussMomentJ_eq_of_lt`), strictly larger than $J_p(0)$ for $v > 0$ (`lt_lintegral_gaussMomentJ`), and $+\infty$ when $\beta v \geq 2$ (`lintegral_gaussMomentJ_eq_top_of_ge`); hence $\mathbb{E}_+[J_p(X)] < \infty \iff \beta v < 2$ (`lintegral_gaussMomentJ_lt_top_iff`). The expected leading coefficient can be infinite although every sample coefficient is finite. Zero `sorry/axiom`.

1 The things formalised

```
theorem integral_rpow_mul_exp_neg_mul_sq (c p) (hc : 0 < c) (hp : 0 < p) :
  s in Ioi 0, s^(p-1) * exp (-c * s^2) = c^(-p/2) * (1/2) * Gamma (p/2)
theorem gaussMomentJ_zero (p) (h) (hp) : gaussMomentJ p 0 = ^(-
p/2) * (1/2) * Gamma (p/2)
theorem gaussTail_eq_of_lt (p v) (h) (hp) (hv : 0 < v) (h : * v < 2) :
  s in Ioi 0, ofReal (s^(p-1) * exp (-*s^2) * exp (v*(s)^2/2))
  = ofReal (gaussMomentJ p 0 * (1 - *v/2)^(-p/2))
theorem lintegral_rpow_Ioi_one_eq_top (p) (hp : 0 < p) : s in Ioi 1, ofReal (s^(p-
1)) =
theorem gaussTail_eq_top_of_ge (p v) (h) (hp) (h : 2 * v) :
  s in Ioi 0, ofReal (...) =
theorem lintegral_gaussMomentJ_eq_of_lt / _eq_top_of_ge (X) (hX) (v : NNReal)
  (hXg : .map X = gaussianReal 0 v) ...
theorem lintegral_gaussMomentJ_lt_top_iff ... :
  ( , ofReal (gaussMomentJ p (X )) ) < * v < 2
theorem lt_lintegral_gaussMomentJ ... (hv : 0 < v) (h : * v < 2) :
  ofReal (gaussMomentJ p 0) < , ofReal (gaussMomentJ p (X ))
```

2 Mathematics

Combining the exponentials, the integrand is $s^{p-1} e^{-cs^2}$ with $c = \beta(1 - \beta v/2)$. For $c > 0$ the Gamma integral gives $c^{-p/2} \Gamma(p/2)/2$, and $J_p(0) = \beta^{-p/2} \Gamma(p/2)/2$, so the ratio is $(1 - \beta v/2)^{-p/2}$, which exceeds 1 when $0 < 1 - \beta v/2 < 1$. For $c \leq 0$ the integrand dominates s^{p-1} on $(1, \infty)$, whose integral diverges for $p > 0$, so the Lebesgue integral is $+\infty$; the extended-real formulation makes this a legitimate value rather than an integrability failure.

3 Paper-facing meaning

rem:pop_vs_emp says the expected empirical coefficient is not the population coefficient. The constant-Gaussian model sharpens this in two ways recorded in the mirror's remark: a Jensen gap ($\mathbb{E}[J_p(X)] > J_p(0)$) and a temperature threshold ($\beta v < 2$) beyond which the expected leading coefficient is infinite. As Astra #12 stressed, this concerns the expected asymptotic coefficient; the fixed- n chart integral on a compact chart has finite expectation in all regimes. The model is a Gaussian constant fluctuation, not the empirical process itself (whose finite- n exponential moments need not exist).

4 Lean notes

- `integral_rpow_mul_exp_neg_mul_rpow` with exponent 2 and `Real.rpow_two` converts to the Monoid power s^2 used in our integrands.
- Non-integrability of s^{p-1} on $(1, \infty)$ is `integrableOn_Ioi_rpow_iff one_pos` (iff $p-1 < -1$); converting to $\int^- = \top$ is `hasFiniteIntegral_iff_ofReal` with `not_lt` and `top_le_iff`, using that `Integrable` is the pair `(AESTronglyMeasurable, HasFiniteIntegral)`.
- Pointwise comparison must be restricted to $s > 1$ (`lintegral_mono_ae` with `ae_restrict_of_forall_mem`); a global `lintegral_mono` fails because `rpow` with negative base is not controlled.
- `Real.one_lt_rpow_of_pos_of_lt_one_of_neg` for the Jensen gap; `ENNReal.ofReal_lt_ofReal_iff` needs positivity of the right argument.